Ko Lih Han

Software Engineer | Applied AI & Backend

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EXPERIENCE

Kahap Inc | Software Developer (Part-time) Aug 2022 - Aug 2026

- Built and maintained a production feedback platform from initial implementation through deployment, centralizing submissions, automated notifications, and admin workflows.
- Shipped and debugged features for a Laravel/Vue.js career platform, resolving production application and database issues.
- Maintained production operations for the career platform, including server migrations/upgrades, Laravel queues/workers/schedulers, SSL renewal, and database troubleshooting.

Pac Com Solution | Full-Stack Developer Intern Jun 2021 - Jan 2022

- Developed and maintained Laravel/Vue.js features for a production POS system.

PROJECTS

Text-to-SQL with Verification & Repair

Python | FastAPI | LangGraph | SQLite | Qwen3.5 9B

- Built a local Text-to-SQL pipeline with schema/value grounding, deterministic SQL verification, read-only execution, and bounded repair.
- On a fixed 100-query BIRD DEV subset with Qwen3.5 9B, verification + bounded repair improved execution accuracy (EX) from 33% to 39% and SQL execution success from 75% to 90%, correcting 6 cases with no accuracy regressions.

Hybrid Retrieval for RAG

Python | FastAPI | BM25 | Qwen3-Embedding-0.6B | RRF

- Built a FastAPI hybrid retriever combining BM25, Qwen embeddings, and RRF, exposing lexical, dense, and fused ranks for each result.
- On a fixed 64-query EnterpriseRAG-Bench slice over 5,189 documents, RRF improved Recall@10 from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742 versus BM25.

Multimodal Video QA Agent

Python | Streamlit | Qwen3.5 4B | FFmpeg

- Built a local video QA agent that chooses between transcript search and frame/clip inspection, shows the evidence used, and declines to answer when evidence is insufficient.
- On a small 12-question test, multimodal tools passed 4/6 visual-required cases versus 0/6 with transcript-only, with no visual calls on transcript-answerable cases in that run.

M.S. Thesis: Context-Aware Workplace Implicit Intent

Llama 3.1 8B | Retrieval | Knowledge Graphs | NLP

- Built role- and context-grounded retrieval over 51,414 posts and 20,455 utterances, separating workplace-role evidence from local context evidence.
- On 401 held-out utterances, combining role and context retrieval improved semantic F1 from 0.6250 to 0.6890.

SKILLS

AI / Retrieval: RAG, Information Retrieval, LLM Evaluation, Agents, Tool Calling, LangGraph
Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js
Databases / Dev Tools: MySQL, SQLite, Linux, Git, GitHub Actions, pytest
Languages: Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | M.S., Information Systems & Applications Sep 2024 - Jun 2026
GPA 4.09/4.3 | Taiwan

Chung Hua University | B.S., Computer Science & Information Engineering Sep 2022 - Jun 2024
GPA 3.9/4.0 | Taiwan